

Date: _____

Day: _____

Blockchain and cryptocurrency

lec 1: (BC)

Blockchain & its application

Golang

Solidity

History $\swarrow$ 1500 BC
Bartering system.
 $\searrow$ Base of accounting

- Commodity money $\rightarrow$ which has some value, goats, cows
- 1st paper money was called IOU (I owe you)

~~Ex: \$100~~

22-01-2023

lec 2:

$\rightarrow$ Distribute $\nearrow$ no single point of failure.

$\rightarrow$ Immutable $\rightarrow$ Base of BC.

$\rightarrow$ cannot be altered. after committed to ledger.

$\rightarrow$ auditable. $\checkmark$

$\rightarrow$ irrevocable $\rightarrow$ if you want to revert/undo perform
also another offset transaction.

$\rightarrow$ Data Integrity $\Rightarrow$ this is why BC is used.

$\rightarrow$ smart contract code $\Rightarrow$ Digital code of contract.

$\rightarrow$ anonymity $\Rightarrow$ all transaction details are available
on blockchain but from whose wallet
the transaction is made is hidden.

$\rightarrow$ only you can perform transaction $\nearrow$ asymmetric transaction.
~~with~~ with public & private key.

$\rightarrow$ making transaction non-reversible.

• Concept of consensus

↳ for transaction validity

↳ done using ledger → everyone has 100% ledger.

• Blockchain is disruptive ~~disruption~~

lec3

27-01-25

• Transaction fees 0.002-0.004

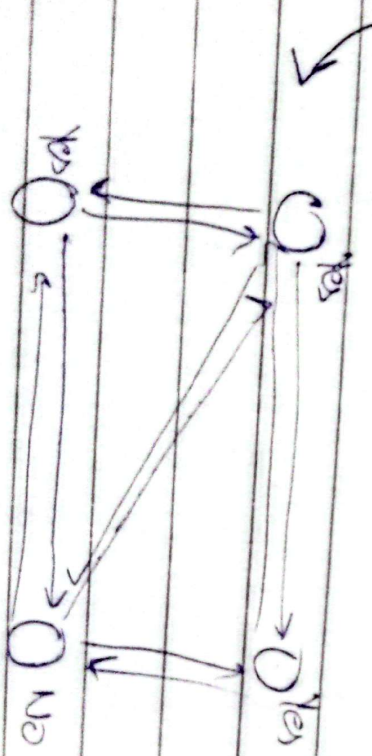
• transaction history available

↳ transparent

↳ Publicly available.

Byzantine General Problem;

↳ message tree



Yes consensus 3/4 = 75%

NO consensus 1/4 = 25%

Ques:

Ans:

Bitcoin Monetary Policy

mining: ↳ The halving

exploring block ↳ Block Frequency

that will be

later added

in term.

↳ Done by miners (are mines)

1 BTC = 100000000

↳ rewarded Bitcoins.

↑

↳ 1 Block = 3.125 Bitcoins.

↳ current.

↳ after 4 years this reward value is

being halved, this is halving

this will ↳ last halved = 2024 = 3×125

end in 2020 2020-2024 = 6×25

plan till 2024 is 2016-2020 = 12×5

available. 2012-2016 = $25 \dots$

↳ rewarding

will end.

• On an average it will take 10 min to create/min a block

↳ need basic machines (simple laptop work)

↳ application specific integrated circuit

• In order to maintain the 10 min average,

he used complexities & hashing

↳ if one block take 5, other will take

15 to balance 10 min average.

• Proof of work → Bitcoin (POW) → Backbone: none

• Proof of stay → Ethereum

Date: _____

Day: _____

Types:

Genesis $\Rightarrow$ 1st Block

Valid $\Rightarrow$ Part of chain

Orphan $\Rightarrow$ Multiple miners mine

Same block at same time,

one of them becomes valid and

other orphan/uncle.

Independent $\Rightarrow$ Pool mining

$\Rightarrow$ different level of hash

$\hookrightarrow$ reward based on hash power.